

Euclid, Book I, Proposition I.8

If Two Triangles Have Three Corresponding Sides Equal, Then Their Corresponding Angles Are Equal

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1 Introduction

Proposition I.8 is one of the central results of Book I of Euclid's *Elements*. It establishes the Side–Side–Side criterion for triangle congruence, proving that triangles having three corresponding sides equal must also have their corresponding angles equal.

Within Euclid's development, this proposition concludes a carefully prepared sequence of results. Proposition I.7 provides the essential uniqueness argument that allows the final congruence theorem to be established.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been derived from the Hilbert axioms before the Euclidean propositions are reconstructed. Consequently, Proposition I.8 becomes a direct application of this general theorem rather than the culmination of a long geometric argument.

This proposition therefore marks the point where the historical proof of Euclid and the modern Hilbert reconstruction differ most significantly while establishing exactly the same mathematical result.

2 The Classical Configuration

Consider two triangles,

$$\triangle ABC \quad \text{and} \quad \triangle DEF,$$

whose corresponding sides satisfy

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

The objective is to prove that the corresponding angles of the two triangles are also congruent.



Figure 1: Two triangles with three corresponding sides congruent.

2.1 Superposition

Euclid conceptually places one triangle upon the other so that the vertex D coincides with A and the side DE lies along the side AB .

Since

$$AB \cong DE,$$

the endpoint E coincides with B .

The remaining vertex F must then lie at the intersection of two circles centered at A and B with radii equal to the corresponding sides of the first triangle.

Proposition I.7 shows that, on a fixed side of the base, such a point is unique. Consequently, the vertex F must coincide with C .

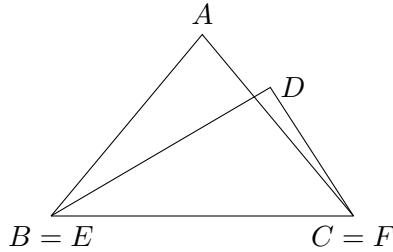


Figure 2: After superposition, the corresponding side $BC = EF$ coincides. If the remaining vertices A and D were distinct, Proposition I.7 would exclude this configuration.

2.2 Application of Proposition I.7

Once the triangles have been superposed, the only remaining possibility is that the third vertices do not coincide.

In that case there would exist two distinct points on the same side of the common base satisfying the same two distance conditions.

Proposition I.7 excludes precisely this situation.

Therefore the remaining vertices must also coincide, and the two triangles agree completely. Consequently, all corresponding angles are congruent.

3 Statement

Theorem 1 (Euclid I.8). *Let*

$$\triangle ABC \quad \text{and} \quad \triangle DEF$$

be nondegenerate triangles.

Assume

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

Then

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

That is, triangles having three corresponding sides congruent also have their corresponding angles congruent.

4 Classical Proof

Superpose the triangle ABC onto the triangle DEF so that the vertex B coincides with E and the side BC lies along the side EF .

Since

$$BC \cong EF,$$

the point C coincides with F .

Suppose that the remaining vertex A does not coincide with D . Then there would exist two distinct points on the same side of the base EF satisfying

$$EA \cong ED$$

and

$$FA \cong FD.$$

This contradicts Proposition I.7, which asserts the uniqueness of such a point.

Therefore the vertex A must coincide with D , and the two triangles coincide completely.

Hence all corresponding angles are congruent:

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This establishes the Side–Side–Side congruence theorem.

5 Proof Architecture of the Reconstruction

The formal reconstruction is dramatically simpler than Euclid’s classical proof.

No superposition argument is required. Once the three corresponding side congruences are available, the previously established Side–Side–Side theorem of the Hilbert interface immediately provides the congruence of the two triangles together with the congruence of all corresponding angles.

The entire classical argument is therefore replaced by a single application of a high-level interface theorem.

6 Hilbert Reconstruction

The Hilbert reconstruction consists of a direct application of the general Side–Side–Side theorem.

The three corresponding side congruences, together with the nondegeneracy of one triangle, satisfy exactly the hypotheses of the derived Hilbert theorem.

Consequently, the theorem immediately yields congruence of the two triangles and, in particular, congruence of all corresponding angles.

Unlike Euclid’s original proof, no superposition argument and no uniqueness reasoning are required. These ideas have already been absorbed into the previously established Hilbert theory.

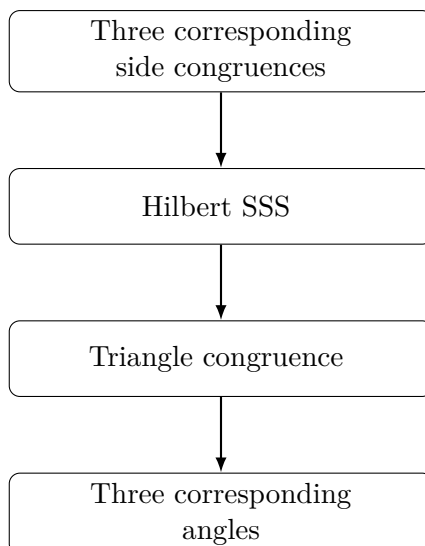


Figure 3: Architecture of the Hilbert reconstruction of Proposition I.8.

7 Lean Formalization

The Lean implementation follows the Hilbert reconstruction directly.

After assuming nondegeneracy of the first triangle and congruence of the three corresponding sides, the proof consists of a single application of the theorem `HilbertSSS`.

This theorem returns a complete triangle congruence result, from which the congruence of the three corresponding angles is obtained by projecting the appropriate fields of the resulting structure.

The formal proof therefore contains no auxiliary constructions, no superposition argument, and no separate uniqueness reasoning. These steps have already been incorporated into the previously established Hilbert interface.

8 Relationship with Earlier Propositions

8.1 Proposition I.4

Proposition I.4 establishes SAS: two sides and the included angle determine the remaining side and angles of a triangle.

Proposition I.8 provides the complementary rigidity theorem. If all three corresponding sides are congruent, then the corresponding angles are congruent as well.

Together I.4 and I.8 provide the two fundamental triangle-congruence principles used repeatedly in the rest of Book I.

8.2 Proposition I.5

The isosceles-triangle theorem supplies an early example of recovering angle information from segment congruence. Proposition I.8 generalizes that phenomenon from the special isosceles configuration to arbitrary triangles with three corresponding sides congruent.

8.3 Proposition I.7

Euclid uses Proposition I.7 to exclude the possibility that two different vertices on the same side of a base could realize the same two prescribed distances from the endpoints of that base.

This uniqueness result is the critical classical ingredient in I.8: after superposition, I.7 forces the remaining vertex to coincide with its counterpart.

8.4 Transition to Proposition I.9

Proposition I.9 uses SSS immediately. Its auxiliary construction creates two triangles with three corresponding sides congruent, and I.8 then yields equality of the two angles forming the desired angle bisector.

Thus I.8 becomes a central reusable congruence theorem as soon as it is established.

9 Hilbert and Book Zero Components Used

9.1 Hilbert SSS

The formal reconstruction packages the SSS rigidity statement as a Hilbert theorem rather than reproducing Euclid’s superposition argument inside every later application.

Its conclusion supplies the corresponding angle congruences required by the public Euclidean proposition.

9.2 Same-Side and Opposite-Side Geometry

The classical proof relies on the phrase “apply one triangle to the other” and on the location of the transported vertex. In the Hilbert setting this positional information is expressed explicitly through side-of-line relations.

9.3 Angle Congruence

The target of I.8 is not merely coincidence of triangles but congruence of corresponding angles. The formal theorem therefore returns the angle-congruence information in the orientation required by subsequent Book I propositions.

9.4 Nondegeneracy

The triangles must be genuine triangles. The formal hypotheses record noncollinearity explicitly, preventing degenerate configurations that are silently excluded by the classical diagram.

10 Formalization Notes

Proposition I.8 concludes the first major block of triangle congruence theory developed in Book I.

Historically, Euclid establishes the Side–Side–Side criterion only after proving the uniqueness result of Proposition I.7 and employing the method of superposition. The classical proof therefore depends on a carefully ordered sequence of earlier propositions.

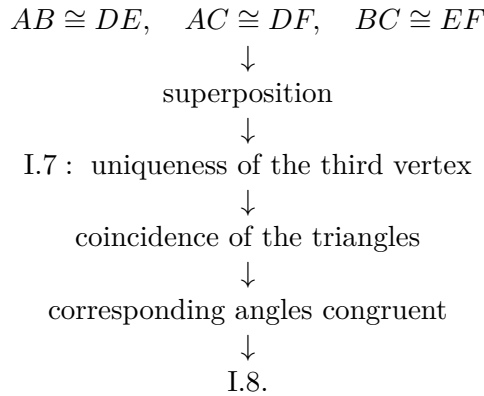
The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been established as a theorem of the Hilbert interface. Consequently, Proposition I.8 becomes a direct application of this previously developed result, requiring no additional geometric construction.

Together, Propositions I.4 through I.8 illustrate the progressive development of triangle congruence theory. The classical arguments of Euclid are systematically replaced by applications of increasingly powerful synthetic theorems derived from the Hilbert axioms.

This transition marks the completion of the first coherent layer of the CGJteamLab Hilbert interface and demonstrates how a modern axiomatic framework can substantially simplify the formal verification of classical geometry.

11 Dependency Summary

The classical dependency structure is



In the Hilbert reconstruction, the same rigidity is isolated as the SSS theorem and used through the congruence interface.

No numerical lengths, coordinates, or new geometric axioms are required by the public Proposition I.8.

12 Conclusion

Proposition I.8 establishes Side-Side-Side congruence for triangles.

Historically, Euclid proves the result by superposition and uses Proposition I.7 to rule out a second possible position of the remaining vertex. The reconstruction makes the underlying rigidity theorem explicit in the Hilbert layer.

This is a major infrastructural point in Book I. From I.8 onward, three segment congruences can be converted into angle congruences without reconstructing the superposition argument each time.

The theorem is used immediately in Proposition I.9 and continues to support midpoint, perpendicular, and later triangle constructions. Thus I.8 is not only a classical proposition but one of the principal congruence interfaces of the formal development.